

Skills Summary

Distance and Midpoint on the Coordinate Plane

Use this reference while completing your worksheet or when studying.

Skill 1 — The distance formula

Where it comes from. Draw the horizontal and vertical gaps between the two points: they are the legs of a right triangle whose hypotenuse is the segment. Pythagoras then gives

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Squaring erases the signs, so it never matters which point you call first.

Example. Find the distance between $(1, 2)$ and $(7, 10)$.

Step 1. Want a straight-line length from two points — so build the right triangle: the legs are the gaps in x and in y , and the distance is its hypotenuse.

Step 2. gaps $7 - 1 = 6$ and $10 - 2 = 8$, so $d = \sqrt{6^2 + 8^2} = \sqrt{36 + 64}$
 $d = 10$ units

Try it. Find the distance between $(-2, 1)$ and $(3, 13)$.

check: 13 units

(!) **Watch out:** adding the two gaps gives how far you would *walk* along the grid, not the straight-line distance. The square root is what turns two legs into one hypotenuse.

Skill 2 — The midpoint formula

Rule. The midpoint is the pair of averages:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

Half the horizontal run pairs with half the vertical rise, so the averaged point lands on the segment, not merely between the endpoints. The answer is an *ordered pair*, and it may have a half in it.

Example. Find the midpoint of $(-3, 8)$ and $(9, -2)$.

Step 1. Want the point half way along — averaging is what "half way" means, and each coordinate is averaged on its own.

Step 2. $\frac{-3 + 9}{2} = 3$ and $\frac{8 + (-2)}{2} = 3$
 $M = (3, 3)$

Try it. Find the midpoint of $(5, -7)$ and $(-1, 3)$.

check: $M = (2, -2)$

Skill 3 — Working back to a missing endpoint

Rule. If M is the midpoint of $\overline{AB}$ and you know A , then each coordinate of B is

$$2 \times (\text{coordinate of } M) - (\text{coordinate of } A),$$

because the midpoint is the average of the two. Do the x -coordinates and the y -coordinates separately, then check by averaging your answer back.

Example. The midpoint of $\overline{AB}$ has x -coordinate 4, and endpoint A has x -coordinate -2 . Find the x -coordinate of B .

Step 1. The midpoint's coordinate is the average of the two endpoints', so doubling it and removing the endpoint I know leaves the one I want.

Step 2. $x_B = 2(4) - (-2) = 8 + 2$

$$x_B = 10$$

Try it. The midpoint of $\overline{AB}$ has x -coordinate 1, and endpoint A has x -coordinate 6. Find the x -coordinate of B .

check: $x_B = 10$